Fiscal Year:	FY 2021	**Task Last Updated:**	FY 02/02/2021
PI Name:	Mao, Xiao Wen M.D.		
Project Title:	Effects of Low-Dose Radiation on Neurovascular Remodeling and Blood-Retina Barrier Function in Mice, Rats and Rabbits		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes (2) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	xmao@llu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	909-558-8373
Organization Name:	Loma Linda University		
PI Address 1:	Radiation Medicine		
PI Address 2:	11175 Campus Street, Rm A1010, Chan Shun Pavilion		
PI Web Page:			
City:	Loma Linda	**State:**	CA
Zip Code:	92350-0001	**Congressional District:**	31
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-Crew Health (FLAGSHIP, OMNIBUS). Appendix A-Omnibus, Appendix B-Flagship
Start Date:	12/14/2017	**End Date:**	09/30/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2022 per NSSC information (Ed., 8/13/21) NOTE: End date changed to 09/30/2021 per NSSC information (Ed., 5/20/21) NOTE: End date changed to 02/28/2021 per NSSC information (Ed., 9/21/20) NOTE: End date changed to 11/30/2020 per NSSC information (Ed., 8/8/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sun, Shu-Wei Ph.D. (Loma Linda University)		
Grant/Contract No.:	80NSSC18K0310		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A recent report shows that more than 50% of the astronauts returning from space were diagnosed with eye problems that can cause headaches and blurry vision. To date, the mechanisms behind these effects are not fully understood. These visual disturbances have been hypothesized to be related to increases in intraocular (IOP) or intracranial pressure (ICP). Modeling study showed that compromise in the integrity of the vascular blood-brain barrier (BBB) would serve to elevate ICP. Damage to microvascular endothelial cells induced by oxidative stress is one of the major factors thought to underlie the increased permeability of the blood-retinal barrier (BRB), which maintains a similar function in the eye for regulating intraocular pressure as the BBB in the cranium. Our knowledge about low-dose radiation risks with long-term spaceflight on retinal vasculature and their potential role in altering the BRB is very limited and inconsistent. The objectives of this application are to characterize the impact of radiation doses and qualities on retinal vasculature and tissue remodeling and to study the role of mitochondria in regulating oxidative stress-induced cellular damage and alteration of cell-cell interactions in the function of blood-retina-barrier (BRB). Archived ocular tissues samples at dose ranges from 0.1 Gy to 1 Gy with multiple particle types and energies across multiple animal species will be available for sharing from three NASA funded investigators. We will evaluate radiation–induced retinal vascular and tissue remodeling at multiple time points. Our study will elucidate the spectrum of radiation-induced structural and functional alteration in retinal BRB function and the mechanisms by which these changes are mediated. Together, our unique, integrative, and quantitative activities with advanced MRI imaging techniques will provide insight into the molecular mechanisms and pathways of low-dose radiation-induced mitochondrial oxidative damage on retinal cell-cell communication, BRB integrity, and vascular remodeling. The data from established model will provide unique criteria for risks of functional detriments associate with spaceflight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward future countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The data from established model will provide unique criteria for risks of functional detriments associated with spaceflight. Our study will also lead to new information to understand the causes and possible treatments of similar Earth-based neurovascular-related diseases and retinal disorders and will point toward future countermeasures.
Task Progress:	Recent study shows that rats were more susceptible to radiation damage in cognitive function when they were sleep perturbed prior to irradiation. We have characterized the impact of low-dose 28Si and 4He radiation on rat retina and retinal blood-retinal barrier (BRB) function and evaluated the effects of sleep fragmentation on radiation-induced retina damage. Our study revealed that exposure to low-dose ionizing radiation induced cellular oxidative damage that may alter retina structure and BRB integrity with dose and radiation quality dependence. Significant changes in retinal endothelial cells occur at doses as low as 1 cGy. These cellular responses may affect visual acuity and impact the potential risk for developing late retinal degeneration. Furthermore, spaceflight condition enhanced radiation-induced apoptosis in the retina. Quantitative assessment indicated that the density of apoptotic cells measured by terminal deoxynucleotidyl transferase dUTP nick-end labeling (TUNEL) assay in the retinal nuclear layer and ganglion cell layer (GCL) was significantly higher in the irradiated groups of 600 Mev Si with dose dependence from 1 cGy to 10 cGy. Rats that were sleep perturbed prior to irradiation had higher apoptotic cellular damage in the retina compared to that of normal sleep condition. The number of TUNEL positive cells were also compared between 600 Mev Si and 400 MEV He. A significant increase of apoptosis in retinal was detected for Si compared to He group ($p<0.05$). Reactive oxygen species (ROS) are involved in lipid peroxidation. Lipid peroxidation was assessed by measuring 4-HNE, an indicative marker of oxidative stress to the retina There were significant increases in the level of 4-HNE immunoreactivity of irradiated group compare to controls ($p<0.05$). Rats that received 600 Mev Si results showed significant increased oxidative damage compared to 400 Mev He radiation exposure at 10 cGy. Sleep perturbed condition did not induce significant increased oxidative damage compared to rats under normal sleep condition. To examine whether space radiation exposure alters endothelial cell junction and adhesion proteins and retinal barrier integrity, the eye tissue expression of the endothelial cell-cell adhesion molecule, platelet endothelial cell adhesion molecule (PECAM-1) and tight junction (TJ) proteins, such as Zonula occludens-1 (ZO-1) were evaluated. In 400 Mev He mice, increased immunoreactivity for PECAM-1 (red) was predominantly present in GCL and inner nuclear layer (INL). PECAM-1 immunoreactivity was significantly higher in the irradiated group that received 600 MevSi and 400 Mev He compared to control group. Furthermore, ZO-1 expression (red) was detected at the GCL and the photoreceptor inner segment (IS) of the ocular tissues. The level of ZO-1 immunoactivity was significantly lower in the 400 Mev He group that received 10 cGy compared to control and 10 cGy 600 Mev Si groups. There were no significant differences between all groups for rats that received 600 MEV Si exposure. There were no significant differences in PECAM-1 and Zo-1 immunoreactivity under normal or sleep perturbation condition. In summary, our study revealed that exposure to low-dose ionizing radiation induced cellular oxidative damage that may alter retina structure and BRB integrity with dose and radiation quality dependence. Significant changes in retinal endothelial cells occur at doses as low as 1 cGy. These cellular responses may affect visual acuity and impact the potential risk for developing late retinal degeneration. Furthermore, spaceflight condition enhanced radiation-induced apoptosis in the retina. The specific mechanism of complex interactions between sleep disturbance and radiation on retinal damage needs to be further investigated. To characterize retinal and blood-retinal barrier biomarkers around prelaminar region of the optic nerve head (ONH) in the flight mice (micro g) eyes and compare to that of habitat ground controls (GC) or 1 g artificial gravity group (micro g+1g) where these mice were in a centrifugal habitat unit for the duration of the flight. 12 mice (male, C57BL/6J) were acclimated for 3 weeks at Kennedy Space Center (KSC) before launch on SpaceX-12 to the International Space Station (ISS) in August 2017. Mice were singly housed on ISS in the Japan Aerospace Exploration Agency (JAXA) Habitat Cage Units (HCUs) in the Cell Biology Experiment Facility (CBEF) in the Kibo module. Six mice were kept in microgravity and 6 were centrifuged at 1g on ISS for the duration of the flight of 30 days. Within 48 hours of landing, the spaceflight mice were sacrificed and eyes were removed for analysis. Quantitative assessment of ocular tissue demonstrated that flight group induced significant apoptosis in the retina close to ONH region compared to all other control groups ($p<0.05$). There were also significant increases in the expression of platelet endothelial cell adhesion molecule-1 (PECAM-1) and decrease in the expression of BRB-related tight junction protein, Zonula occludens-1 (ZO-1) around ONH region. These changes were more prominent near the ONH region than that of other area of the retina. Less changes were observed in centrifuged group (micro g+1g) compared to micro g group. These data indicate

	profound cellular response of retina structure and BRB function near optic nerve following long-term spaceflight. Changes in retina and retinal vascular property may affect and contribute to optic disc edema due to weightless condition. Our study also demonstrated that application of 1g during spaceflight could mitigate some detrimental effects of microgravity on the retina.
Bibliography Type:	Description: (Last Updated: 07/10/2026)
Abstracts for Journals and Proceedings	Mao XM, Nishiyama N, Stanbouly S, Britten R. "Prolonged exposure of low-dose neutron irradiation impacts on rat blood-retina barrier integrity and ocular structure." 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. Abstracts. 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. , Jan-2020
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Abstracts for Journals and Proceedings	Wang W, Gasheva O, Narayanan A, Zawieja S, Castorena J, Gashey A, Bagher P, Mao XW, Willey J, Delp M, Cromer W, Zawieja D. "Analysis of the structure of mesenteric lymph nodes from space flown rodents." 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. Abstracts. 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. , Jan-2020
Abstracts for Journals and Proceedings	Mao XW, Stanbouly S, Nishiyama N, Delp M. "Spaceflight induces mouse retina changes near optic nerve head." Presented at the 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. Abstracts. 36th Annual Meeting of the American Society for Gravitational and Space Research, Virtual Meeting, November 5-6, 2020. , Nov-2020